Multimodal Deep Learning

Exercise Sheet 8

Date: 8th of July, 2025

Exercise 1: Exploring Diffusion Models in Practice

This tutorial introduces you to the practical application of diffusion models, serving as a hands-on complement to the theory that will be covered in next week's lecture. Please work through the exercises by following the instructions provided in each Jupyter notebook. We will discuss the solutions and your findings in next week's tutorial session.

Computational Resources: This exercise is designed to run on the free tier of Google Colab. This ensures you can complete all tasks even if you do not have access to a local machine with a powerful GPU.

How to Use Google Colab: To run the notebooks, please follow these steps:

- Navigate to `colab.research.google.com` and sign in with your Google account.
- Upload the exercise notebook (`.ipynb` file) by selecting `File > Upload notebook`.
- Ideally, enable GPU acceleration.

(a) First, familiarize yourself with the fundamental unconditional diffusion process by following the instructions in the `00_UnconditionalDiffusion.ipynb` notebook. *Note: This part requires the pre-trained weights file `trained_diffusion_model_large.pth`.*

(b) Next, you will generate a short video clip from a text prompt. Complete the exercises in the `01_VideoGenTutorial.ipynb` notebook.

(c) Now, generate a corresponding audio track using a Text-to-Speech (TTS) model. Work through the tasks in the `02_TTS.ipynb` notebook. *A research paper is linked within the notebook for your interest, but reading it is not required for this exercise.*

(d) Finally, combine the assets you have created. Use the `03_LipSync.ipynb` notebook to synchronize the lip movements in your generated video with the audio from the previous step.

Exercise 2: Warm-up: The Evidence Lower Bound

In many modern generative models, our goal is to maximize the log-likelihood of observed data, $\log p_\theta(\mathbf{x})$, which is often intractable. Variational Inference (VI) reformulates this problem by maximizing a tractable, parameterized lower bound instead. This exercise will guide you through the derivation of this bound (the Evidence Lower Bound (ELBO)) and reveal its relationship with the KL divergence.

We consider a generative model with parameters θ that defines a distribution over observed data $\mathbf{x}$ and latent variables $\mathbf{z}$. The model assumes the latents generate the data, represented as $\mathbf{z} \to \mathbf{x}$.

- **Model Joint Distribution:** $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z}) p_\theta(\mathbf{z})$.
- **True Posterior:** $p_\theta(\mathbf{z}|\mathbf{x}) = \frac{p_\theta(\mathbf{x},\mathbf{z})}{p_\theta(\mathbf{x})}$. This is the "true" but typically intractable distribution of the latents given the data.

- **Variational Posterior:** $q_\phi(\mathbf{z}|\mathbf{x})$. This is our tractable, parameterized approximation to the true posterior, with parameters ϕ. In the context of VAEs, this is often called the *encoder*.
- **KL Divergence:** $D_{\mathrm{KL}}(Q\|P) = \mathbb{E}_{z\sim Q}\left[\log \frac{Q(\mathbf{z})}{P(\mathbf{z})}\right]$. A non-negative measure of dissimilarity, where $D_{\mathrm{KL}}(Q\|P) \geq 0$.

(a) **From KL Divergence to the ELBO** Start with the KL divergence of the variational posterior $q_\phi(\mathbf{z}|\mathbf{x})$ to the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$. Your task is to show that the log-likelihood of the data can be decomposed as follows:

$$\log p_\theta(\mathbf{x}) = D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z} \mid \mathbf{x})\big) + \mathcal{L}(\theta, \phi)$$

where the Evidence Lower Bound, $\mathcal{L}(\theta, \phi)$, is defined as:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$$

Hint: Expand the KL divergence term. Since $\log p_\theta(\mathbf{x})$ *does not depend on* $\mathbf{z}$*, it can be moved outside the expectation over* q_ϕ*.*

(b) **Why Maximize the ELBO?** Using the equation derived in Part A, explain why maximizing the ELBO, $\mathcal{L}(\theta, \phi)$, with respect to both model parameters θ and parameters ϕ is a valid objective. Then, state the condition under which the ELBO becomes tight (i.e., when does $\mathcal{L}(\theta, \phi) = \log p_\theta(\mathbf{x})$?).

(c) **The Reconstruction + Regularization Form** The ELBO is often optimized in a more intuitive form. Using the product rule on the joint distribution, $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})$, show that $\mathcal{L}(\theta, \phi)$ can be rewritten as:

$$\mathcal{L}(\theta, \phi) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{Reconstruction Term}} - \underbrace{D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z})\big)}_{\text{Regularization Term}}$$

This final form reveals the objective we actually optimize in models like Variational Autoencoders (VAEs), balancing data reconstruction with keeping the approximate posterior close to the prior.

Exercise 3: KL Divergence to Mean Squared Error: Simplifying a Generative Objective

In this exercise, you will learn to simplify a complex objective function, a common problem in machine learning. You will use this next week in the lecture when you learn about the diffusion forward and reverse process. You will begin with the Kullback-Leibler (KL) divergence between two Gaussian distributions. Through a series of algebraic manipulations and re-parameterizations, you will prove that minimizing this divergence is equivalent to minimizing a much simpler mean squared error.

Problem Setup (Givens)

For this exercise, you are given the following mathematical objects and relationships. Treat them as axioms.

1. **Constants:** For a time index $t \in \{1, ..., T\}$:

- $\beta_t \in (0, 1)$
- $\alpha_t = 1 - \beta_t$
- $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$

2. **Vector Construction:** A vector $\mathbf{x}_t \in \mathbb{R}^D$ is constructed from a source vector $\mathbf{x}_0 \in \mathbb{R}^D$ and a standard Gaussian noise vector $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}^D)$ according to the following equation:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

3. **Target Distribution (True Posterior):** The true posterior distribution q of $\mathbf{x}_{t-1} \in \mathbb{R}^D$ conditioned on $\mathbf{x}_t$ and $\mathbf{x}_0$ is a Gaussian:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \Sigma(t)\right)$$

where:

- Mean: $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$
- Covariance: The covariance is $\Sigma(t) = \frac{(1-\bar{\alpha}_{t-1})(1-\alpha_t)}{1-\bar{\alpha}_t}$.

4. **Model Distribution (Learned Reverse Process):** We define a model p_θ with learnable parameters θ that follow a Gaussian distribution, intended to approximate q:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I}^D)$$

5. **KL Divergence Formula:** The KL divergence of two D-dimensional diagonal Gaussians, $P_1 = \mathcal{N}(\mu_1, \sigma_1^2\mathbf{I}^D)$ and $P_2 = \mathcal{N}(\mu_2, \sigma_2^2\mathbf{I}^D)$, is:

$$D_{\mathrm{KL}}(P_1\|P_2) = \frac{1}{2}\sum_{i=1}^{D}\left(\log\frac{\sigma_2^2}{\sigma_1^2} - 1 + \frac{\sigma_1^2}{\sigma_2^2} + \frac{(\mu_{1,i} - \mu_{2,i})^2}{\sigma_2^2}\right)$$

(a) **Simplifying the Objective via Fixed Variance**

Our goal is to optimize the model parameters θ by minimizing the KL divergence of the target distribution q to the model distribution p_θ. The objective function for a given time step t is:

$$L_t(\theta) = D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

To simplify the problem, we make a crucial modeling choice: we fix the variance of our model p_θ to be equal to the variance of the target q. That is, we set the model's variance σ_t^2 to the known value of the variance of the variational posterior, $\Sigma(t)$.

Your Task: Show that with the choice $\sigma_t^2 = \Sigma(t)$, minimizing $L_t(\theta)$ with respect to θ is equivalent to minimizing the squared Euclidean distance between the means of the two distributions:

$$\min_\theta L_t(\theta) \iff \min_\theta \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2$$

Hint: The parameters θ only influence μ_θ. Identify all terms in the KL divergence formula that do not depend on θ and can thus be treated as constants during optimization.

(b) **Re-parameterizing the Target Mean**

The expression for the target mean $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$ depends on $\mathbf{x}_0$, which is unavailable during the reverse process. We can re-express this mean entirely in terms of the known $\mathbf{x}_t$ and the noise vector ϵ that generated it.

Your Task: Start by isolating $\boldsymbol{x}_0$ from the $\boldsymbol{x}_t$ definition (Given #2). Then substitute this into the expression for $\tilde{\mu}_t$ (Given #3) and group the terms for $\boldsymbol{x}_t$ and ϵ. Through algebraic simplification, prove that the target mean can be rewritten as:

$$\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right)$$

Hint: This is a direct algebraic manipulation. You will need to use the identities $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$ *and* $\beta_t = 1 - \alpha_t$ *to simplify the coefficients of the* $\mathbf{x}_t$ *and* ϵ *terms.*

(c) **Deriving the Final, Practical Objective**

We now use the insight from Part B to re-parameterize our model's mean. Instead of designing a network to predict the complex mean μ_θ directly, we will design a network $\epsilon_\theta(\mathbf{x}_t, t)$ that attempts to predict the noise vector ϵ from the noisy vector $\mathbf{x}_t$. The model's mean is then constructed using the same functional form as the true mean:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$$

Your Task: Substitute this new definition for μ_θ and the simplified expression for $\tilde{\mu}_t$ from Part B into the simplified objective from Part A. Show that this yields an objective proportional to the mean squared error between the true noise ϵ and the predicted noise ϵ_θ. Specifically, show that:

$$\|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2 = C \cdot \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2$$

where C is a positive constant that does not depend on the model parameters θ. This proves that training the model is equivalent to a simple noise-prediction task. Determine the expression for C in terms of α_t, β_t, and $\bar{\alpha}_t$.

This final result proves that minimizing the original, complex KL divergence objective is equivalent to training a neural network on a simple noise-prediction task using a standard MSE loss. After removing the constant weighting term C, we arrive at the famous L_{simple} objective from the DDPM paper `https://arxiv.org/pdf/2006.11239`

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2 \right]$$